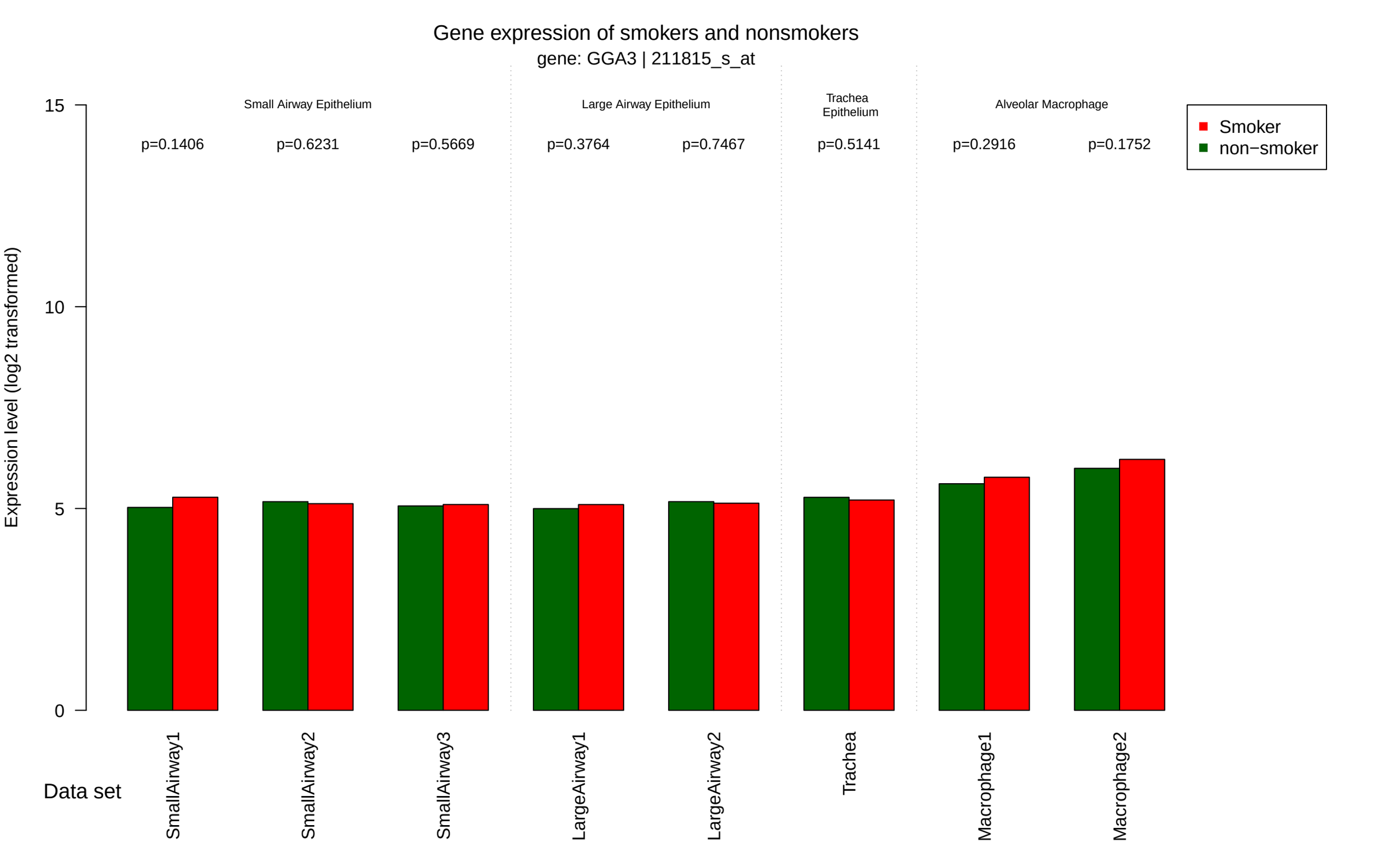
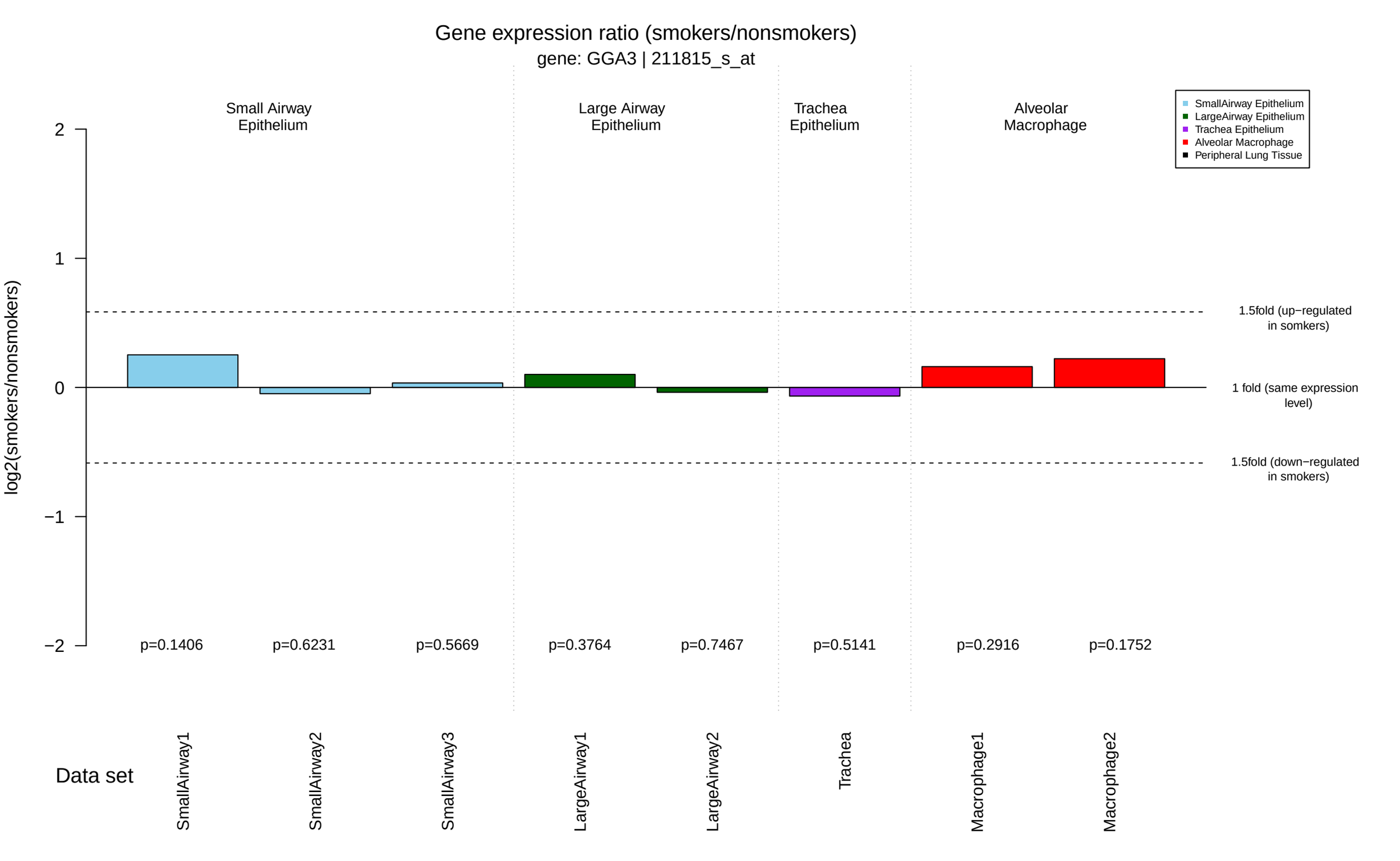
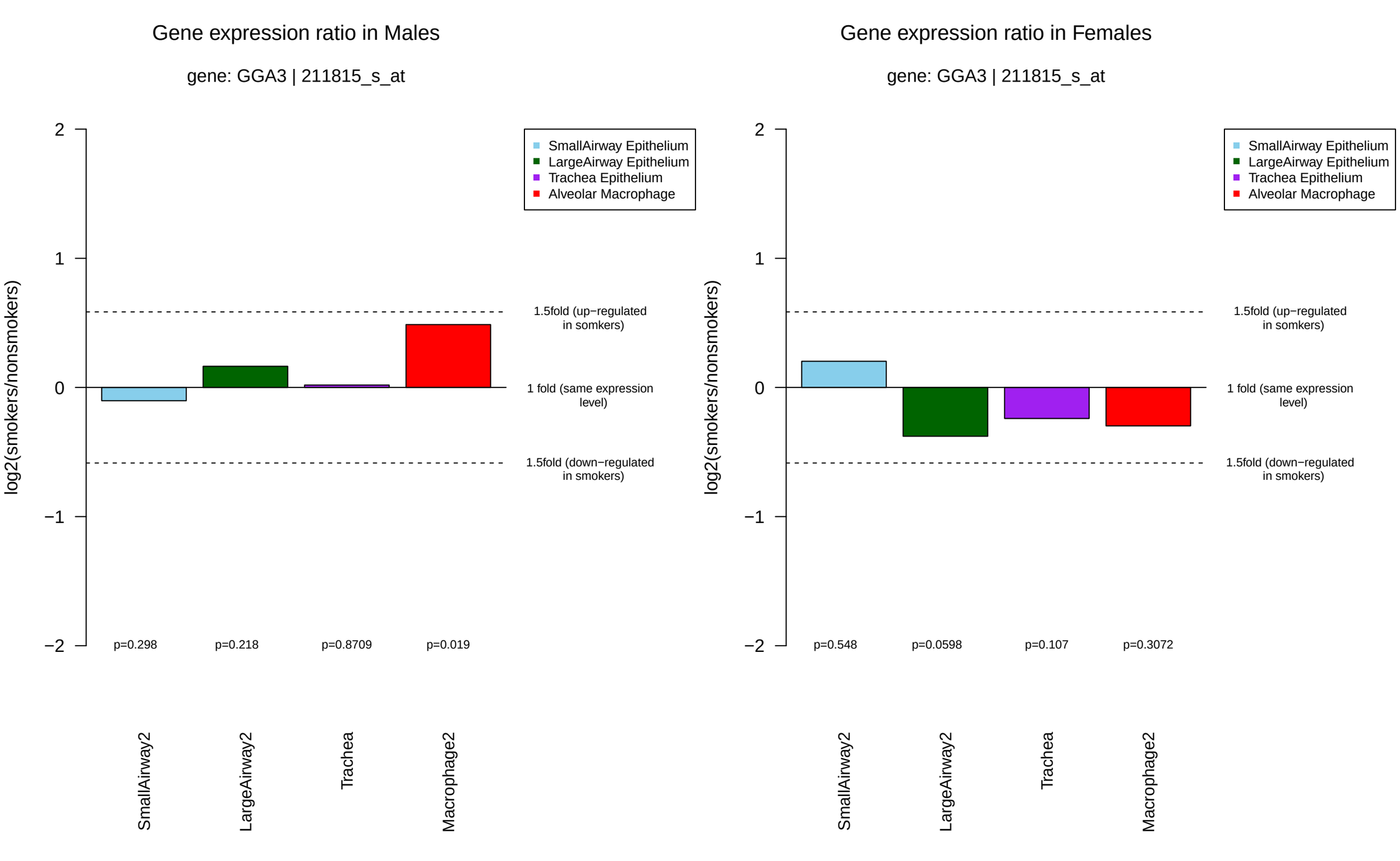
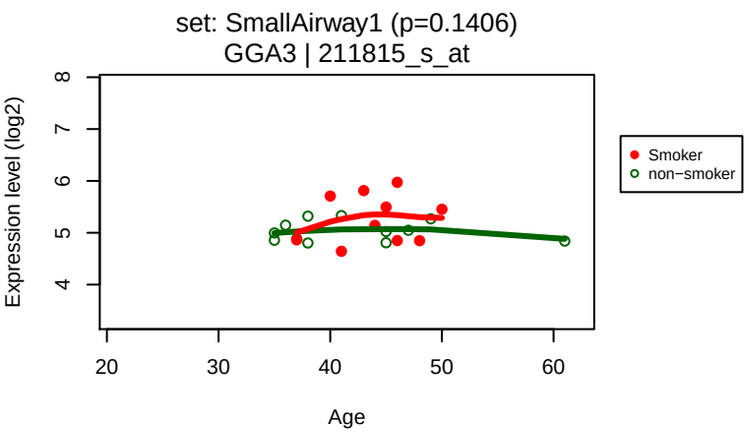
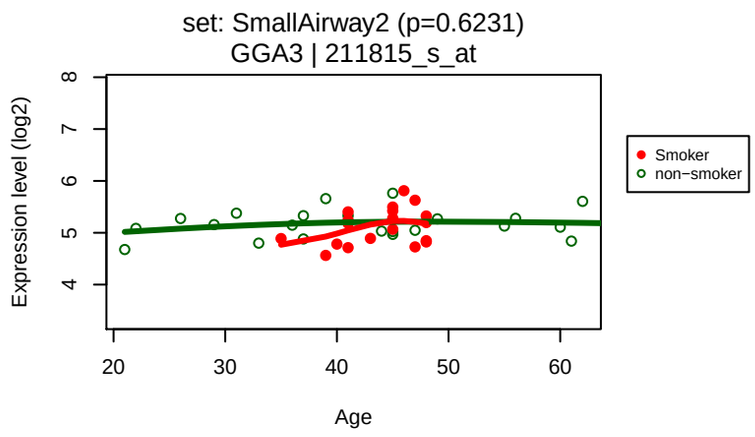
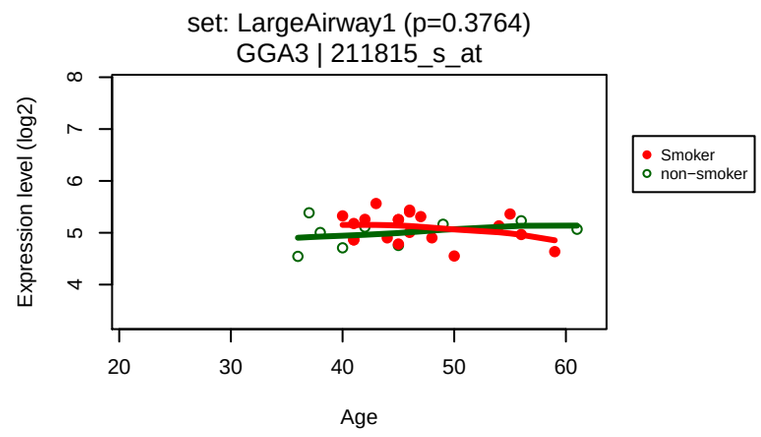

Gene expression of smokers and nonsmokers
gene: GGA3 | 211815_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1406
p=0.6231
p=0.5669
p=0.3764
p=0.7467
p=0.5141
p=0.2916
p=0.1752
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

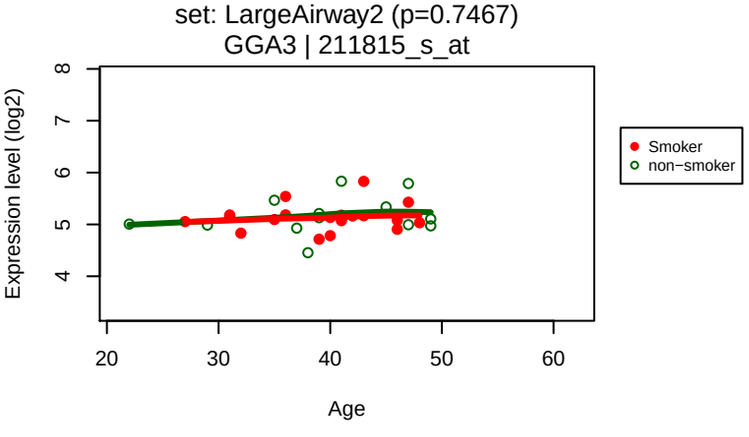

Gene expression ratio (smokers/nonsmokers)
gene: GGA3 | 211815_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1406
p=0.6231
p=0.5669
p=0.3764
p=0.7467
p=0.5141
p=0.2916
p=0.1752
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

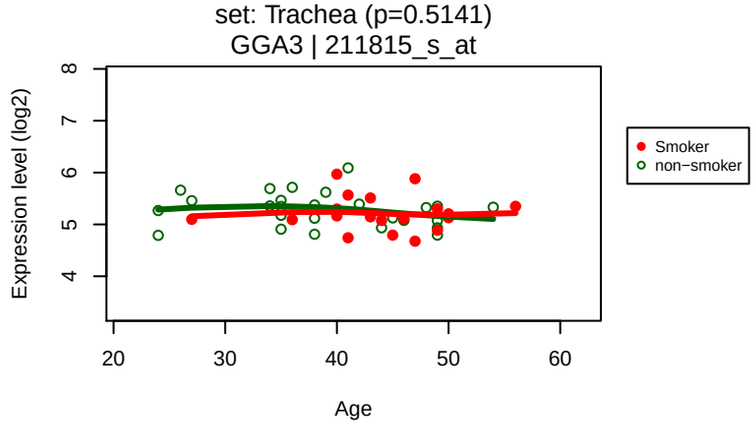

Gene expression ratio in Males
gene: GGA3 | 211815_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.298
p=0.218
p=0.8709
p=0.019
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GGA3 | 211815_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.548
p=0.0598
p=0.107
p=0.3072
SmallAirway2
LargeAirway2
Trachea
Macrophage2

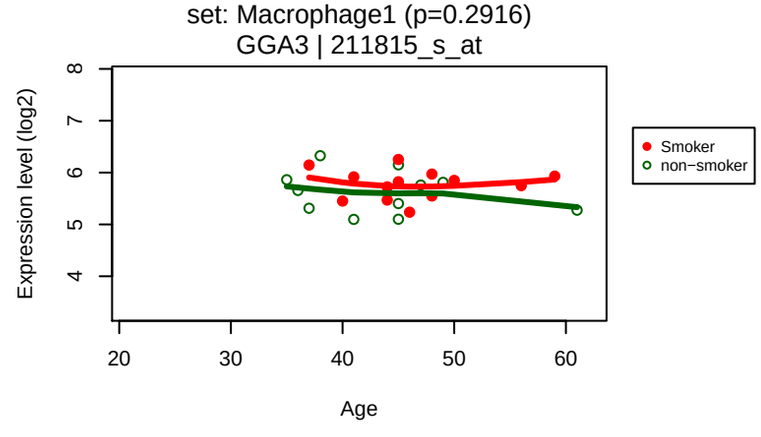

set: SmallAirway1 (p=0.1406)
GGA3 | 211815_s_at
Expression level (log2)
Age
Smoker
non−smoker

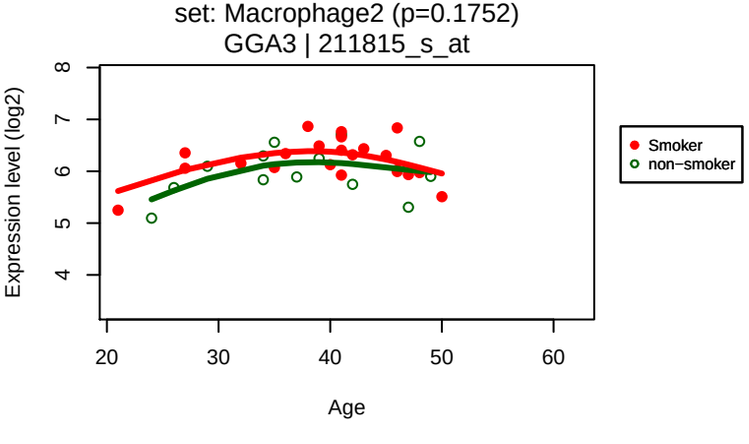

set: SmallAirway2 (p=0.6231)
GGA3 | 211815_s_at
Expression level (log2)
Age
Smoker
non−smoker

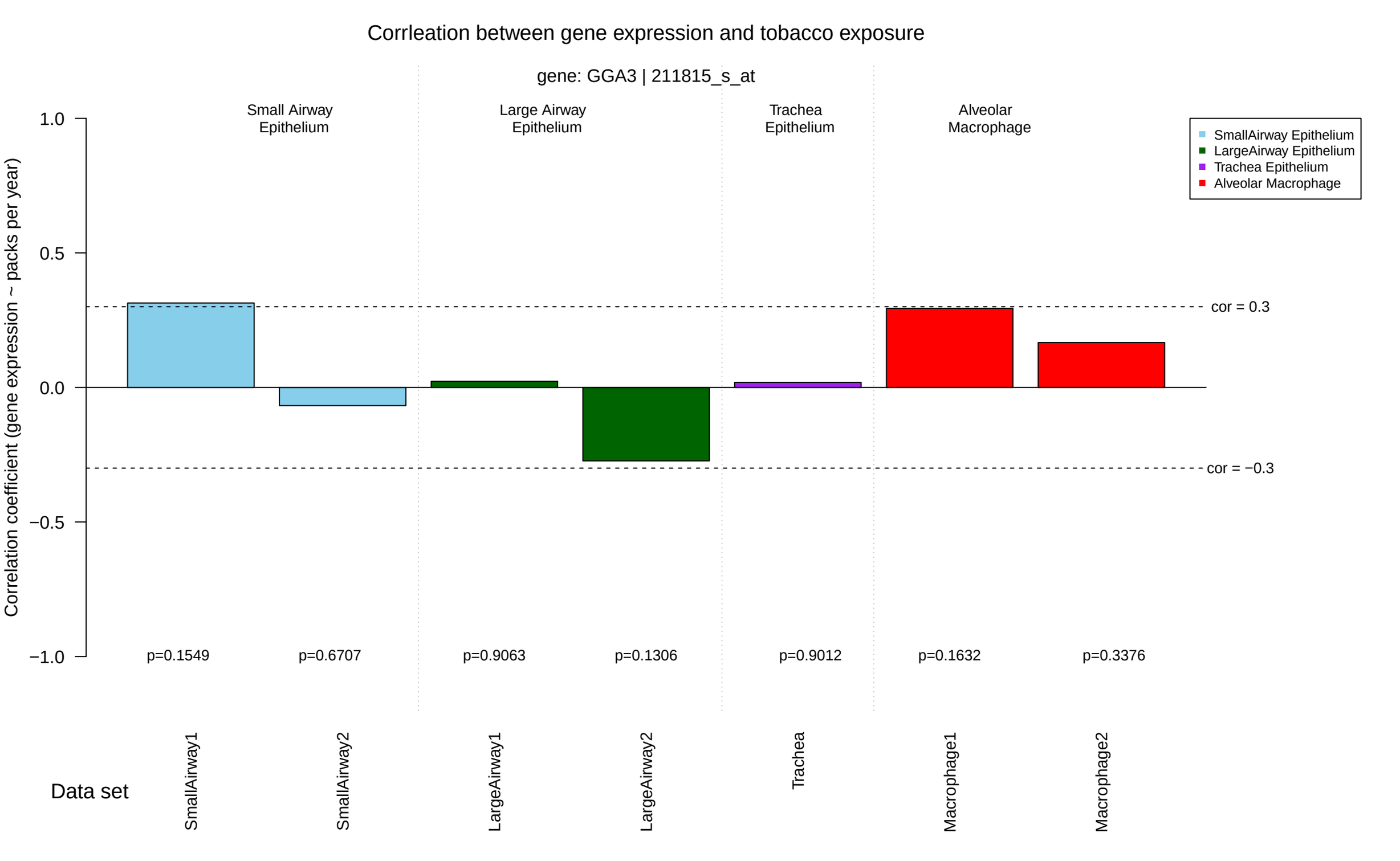

set: LargeAirway1 (p=0.3764)
GGA3 | 211815_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7467)
GGA3 | 211815_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5141)
GGA3 | 211815_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2916)
GGA3 | 211815_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1752)
GGA3 | 211815_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GGA3 | 211815_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1549
p=0.6707
p=0.9063
p=0.1306
p=0.9012
p=0.1632
p=0.3376
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2